Probabilistic forecasting of UK electricity demand: quantile regression vs machine learning

A short research project extending my MTech thesis method (linear quantile regression, FUTA, 2018) to modern machine learning, applied to half-hourly UK electricity demand.

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Status: initial study complete (one year of data, single time-ordered split). Further tests are listed under Next steps.

Research question

Do machine learning quantile models give better calibrated and sharper prediction intervals for half-hourly UK electricity demand than classical linear quantile regression?

Data

National Energy System Operator (NESO) "Historic Demand Data" for 2025, half-hourly. Columns used: `SETTLEMENT_DATE`, `SETTLEMENT_PERIOD`, `ND` (national demand). Source: NESO data portal (search "Historic Demand Data"). The raw file is **not** included in this repository. Please download it from NESO and check its open data licence for the reuse terms. Data: NESO.

After cleaning, 17,518 half-hourly observations were used (13,745 for training and 3,437 for testing).

Models compared (each at quantiles 0.1, 0.5 and 0.9)

1. Linear quantile regression (statsmodels): the thesis baseline
2. Gradient boosting with quantile loss (scikit-learn)
3. Random forest, with quantiles taken from the spread of tree predictions (scikit-learn). This is a simple approximation and tends to produce intervals that are too narrow.

Features: time of day and day of year (sine and cosine), weekday and weekend, demand 24 hours earlier and demand 7 days earlier.

Evaluation

- Pinball loss at each quantile (lower is better)
- Coverage of the 80% prediction interval (target 0.80)

- Mean width of the 80% interval
- Time-ordered split: the first 80% of the year for training, the final 20% (roughly late October to December) for testing, with no shuffling

Results

Model	Pinball 0.1	Pinball 0.5	Pinball 0.9	Coverage (80%)	Mean width (MW)	Avg pinball
Linear quantile regression	381.2	813.2	365.2	0.772	5157	519.9
Gradient boosting	621.4	896.4	347.9	0.548	3150	621.9
Random forest	440.6	805.7	397.9	0.525	3142	548.1

The linear quantile regression had the lowest average pinball loss and the best coverage (77.2%, close to the 80% target). The machine learning models gave narrower intervals but covered the observed demand only 52% to 55% of the time, so they were overconfident.

Interpretation (not yet tested). One possible reason is that the test period contains late-autumn and winter demand levels that the models did not see during training, and tree-based models cannot extrapolate beyond their training range. This study cannot confirm that. It uses a single year and one split, so it does not show that machine learning is worse than quantile regression in general.

How to run

1. Open `demand_quantile_forecasting.py` in Google Colab, or run it locally with Python 3.10+ and `pandas`, `numpy`, `scikit-learn`, `statsmodels`, `matplotlib`.
2. Put the NESO CSV in the same folder and set `DATA_PATH` at the top of the script.
3. Run the script. It writes `metrics.csv` and three charts to a `results/` folder.

Limitations

- One year of data and a single train and test split.
- Calendar and lagged demand only; no weather and no bank holidays.
- Random forest intervals are approximate.
- One country.

Next steps

- Hold out one week from each month, and train on a second year, to test the explanation above
- Add bank holiday and weather features
- Use a proper quantile regression forest and conformal calibration of the intervals
- Test deep learning (for example an LSTM trained with pinball loss)

Acknowledgements

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Licence

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